

Second-Harmonic Generation with a 440 000% W⁻¹ Conversion Efficiency in a Lithium Niobate Microcavity without Periodic Poling

Xiao Wu, Zhenzhong Hao, Li Zhang, Di Jia, Rui Ma, Can Tao, Feng Gao, Fang Bo,*
Guoquan Zhang, and Jingjun Xu

Thin-film lithium niobate (TFLN) enables extremely high-efficiency second-order nonlinear optical effects due to large nonlinear coefficient d_{33} and strong optical field confinement. Here, a pulley-waveguide-coupled microring resonator with an intrinsic quality factor above 9.4×10^5 on the reverse-polarized double-layer X-cut TFLN is designed and fabricated. In such a TFLN resonator without fine domain structures, second harmonic generation with a normalized conversion efficiency of 440,000% W⁻¹ with weak pump power, approaching that in periodically poled lithium niobate (PPLN) microring resonators, and an absolute conversion efficiency of 30% is realized with a sub-milliwatt continuous pump. This work reduces the dependence of high-efficiency nonlinear frequency conversion on PPLN microcavities that are difficult to prepare.

realized a high conversion efficiency of 5540% W⁻¹ cm⁻² in dual-layer TFLN waveguides,^[16] comparable to the SH conversion efficiency of PPLN waveguides.^[17,18] The main advantages of MPM on RP TFLN compared to PPLN are: 1) The effective nonlinearity coefficient is equal to d_{33} instead of $2/\pi \cdot d_{33}$, although at the cost of smaller mode overlap, which results in a two-fold increase in the normalized SH conversion efficiency in the waveguides. 2) Electrode preparation and subsequent rigorous periodic poling processes are not required.

Here, we designed and prepared pulley-waveguide-coupled racetrack microresonators on an RP double-layer X-cut TFLN. In these TFLN resonators without periodic domain structures, the effective nonlinear coefficient is 14 pm V⁻¹ through a Fourier transform of nonlinear coefficient in the whole microcavity. An ultra-high normalized conversion efficiency of 440000% W⁻¹ was achieved under nearly critical pump coupling and dual resonance in the fundamental-harmonic (FH) and SH bands. The normalized SH conversion efficiency exceeds that of all photonic microcavities based on MPM, only an order of magnitude lower than PPLN microcavities with complex preparation processes.^[19] Furthermore, a 30% absolute conversion efficiency was observed with a 600 μ W on-chip pump power.

1. Introduction

Ultra-efficient second harmonic (SH) generation based on second-order nonlinearity has essential applications in classical light sources^[1–3] and quantum light sources.^[4–6] Thin-film lithium niobate (TFLN) microring resonators with high-quality factors have been widely used in nonlinear photonic integration platforms due to their large nonlinear coefficient ($d_{33} = 25$ pm V⁻¹), wide transparent window (0.4–5 μ m) and tight light field confinement.^[7–11] The flexible microdomain processing of TFLN enables quasi-phase-matched periodically poled lithium niobate (PPLN) photonic waveguides and microcavities with extremely high nonlinear optical conversion efficiencies.^[12–14] In 2020, Zhang et al. proposed a novel mode-phase-matching (MPM) method for SH generation from TE_{00, FH} to TE_{01, SH} mode in reverse-polarized (RP) dual-layer TFLN waveguides, resulting in a theoretical normalized conversion efficiency as high as 13000% W⁻¹ cm⁻².^[15] In 2021, Wang et al.

2. Principles of Device Design and Fabrication

The racetrack resonators were fabricated on an RP double-layer TFLN on a 2 μ m thick silicon dioxide film and 500 μ m thick silicon substrate from NANOLN. The total thickness of the TFLN is 600 nm, and the upper layer has a thickness of 310 nm. The preparation method of TFLN microring resonators is similar to that in Reference.^[20] It mainly includes electron beam lithography to define the etching mask and inductively coupled plasma reactive ion etching to transfer the mask pattern to the TFLN, where the complex electrical or optical poling processes to prepare the microdomain structure of PPLN devices is excluded. The scanning electron microscope (SEM) of a fabricated TFLN microring resonator is shown in Figure 1a. The length of the straight arm of the racetrack resonator is 300 μ m. To reduce mode coupling between the TE_{01, SH} mode, and other high-order modes

X. Wu, Z. Hao, L. Zhang, D. Jia, R. Ma, C. Tao, F. Gao, F. Bo, G. Zhang, J. Xu
MOE Key Laboratory of Weak-Light Nonlinear Photonics
TEDA Institute of Applied Physics and School of Physics
Nankai University
Tianjin 300457, China
E-mail: bofang@nankai.edu.cn

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/lpor.202300951>

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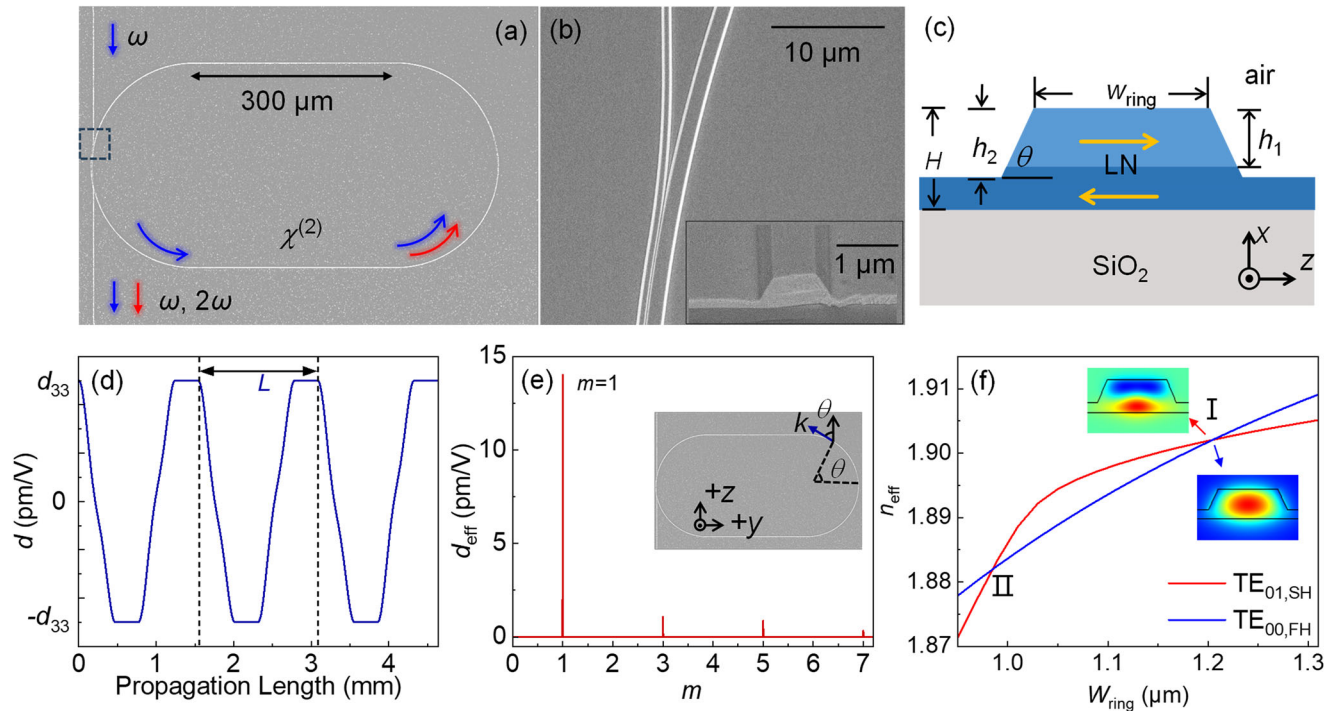


Figure 1. Design and characterization of double-layer TFLN racetrack resonators. a) SEM images of the overall structure of a typical TFLN racetrack ring. b) SEM image of the coupled region between a pulley waveguide and a resonator. The inset is the SEM image demonstrating the end facet of the pulley waveguide. c) Schematic diagram of the cross-section of the straight arm in a racetrack RP dual-layer resonator, where $H = 600$ nm, $h_1 = 310$ nm, $h_2 = 420$ nm, $\theta = 65^\circ$. d) The variation of the nonlinear coefficient d with the propagation length. e) The Fourier transform d_{eff} of the nonlinear coefficients d . The inset is the schematic diagram of the change of angle θ between wave vector k and $+z$. f) The effective refractive indexes of $\text{TE}_{00,\text{FH}}$ and $\text{TE}_{01,\text{SH}}$ modes vary with the microring width in the straight arm. The mode overlap is $0.83 \mu\text{m}^{-1}$ at point I and $0.32 \mu\text{m}^{-1}$ at point II. The insets show the E_z component of the corresponding modes.

ensuring the **adiabatic mode evolution** in the bend section, the **bending radius of the racetrack cavity was set to be 150 μm** . We used a single pulley waveguide to couple and extract the pump and the signal. The width of the pulley waveguide is 450 nm, the coupling gap is 300 nm, and the coupling length is 52 μm . The zoomed SEM images of the coupling part and the cross-section are shown in Figure 1b and its inset. The smooth waveguide surface and side walls benefit from our high-quality fabrication process excluding periodic poling. On the cross-section, the dividing line of the RP double-layer film is faintly visible. Figure 1c schematically illustrates the cross-section of the straight arm of the RP double-layer TFLN resonator, with crystal orientation as coordinates. The yellow arrows in the waveguide indicate the direction of spontaneous polarization of lithium niobate crystal. The etching depth of 420 nm allows $\text{TE}_{01,\text{SH}}$ a minor mode loss (See the **Supporting Information**).

In an X-cut microcavity, the angle θ between the wave vector k and the optical axis z changes as shown in the inset of Figure 1e, which leads to a nonlinear coefficient d changing with the propagation angle θ ,^[21]

$$d = -d_{22}\cos^3\theta + 3d_{31}\cos^2\theta\sin\theta + d_{33}\sin^3\theta \quad (1)$$

Figure 1d shows the variation of the nonlinear coefficient d with the propagation length. It is seen that the effective nonlinear coefficient d oscillates periodically between $+d_{33}$ and $-d_{33}$ with

a spatial period equal to the cavity length L . It is analogous to the PPLN with a poling period of L . The Fourier transform of d is shown in Figure 1e. It is seen that when $m = 1$, the effective Fourier coefficient d_{eff} is the largest $(d_{\text{eff}})_{\text{max}} = 14.0 \text{ pm V}^{-1}$. For a cavity with $m = 1$, the quasi-phase matching (QPM) condition can be expressed as

$$|2m_{\text{FH}} - m_{\text{SH}}| = 1 \quad (2)$$

where m_{FH} and m_{SH} correspond to the azimuthal numbers of pump and SH light, respectively. The azimuthal number follows the resonance condition

$$n_{\text{FH,SH}} \times L = m_{\text{FH,SH}} \times \lambda_{\text{FH,SH}} \quad (3)$$

where $n_{\text{FH,SH}}$ are the effective indices for the FH and SH modes, respectively, and L is the total perimeter of the cavity. $\lambda_{\text{FH,SH}}$ are the wavelengths of FH and SH light. From Equations (2) and (3), the relations between n_{FH} and n_{SH} can be obtained,^[22]

$$n_{\text{SH}} = \left(1 \pm \frac{1}{2m_{\text{FH}}}\right) n_{\text{FH}} \quad (4)$$

In this work, the perimeter of the resonator we used is 1542 μm , which corresponds to an m_{FH} value of around 1912 according to $m_{\text{FH}} = \frac{1}{\lambda_{\text{FH}}} \int_0^L n_{\text{FH}}(l) dl$. In this case, $1/(2m_{\text{FH}})$ is much

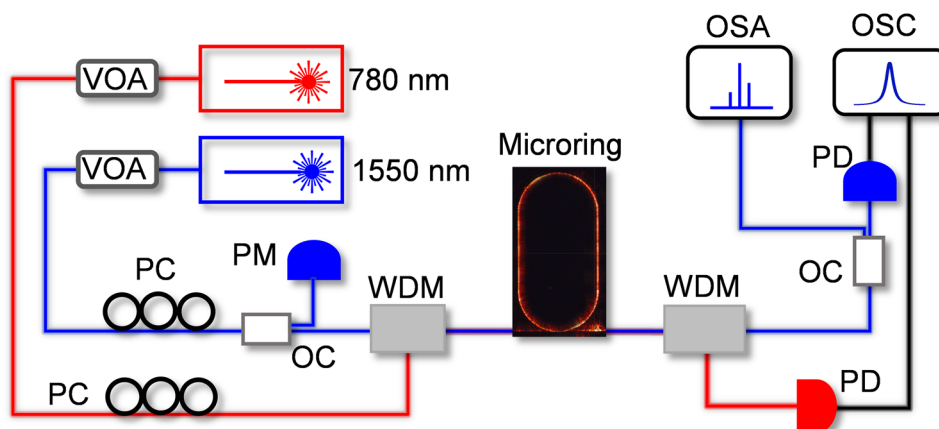


Figure 2. The experimental setup for SH detection. The microring is indicated by a photograph of a racetrack resonator scattering the SH signal. VOA, variable optical attenuator; PC, polarization controller; WDM, wavelength division multiplexer; OC, optical coupler; PM, power meter; PD, photodetector; OSC, oscilloscope; OSA, optical spectrum analyzer.

lower than 1, so we can design the waveguide geometry to fulfill the phase-matching conditions similar to a straight waveguide.

The phase matching condition is realized by adjusting the top width of the microcavity. We simulated the effective refractive indices n_{eff} of $\text{TE}_{00, \text{FH}}$ mode around 1550 nm and $\text{TE}_{01, \text{SH}}$ mode near 775 nm in a straight waveguide to simplify the QPM design (Figure 1f). The phase matching condition requires that the n_{eff} of the two modes be approximately equal. Accordingly, the waveguide width is selected to be 1.21 μm , indicating by cross point I of the red and blue lines of Figure 1f, for the larger mode overlap. We prepared waveguides of different widths with steps of 30 nm to compensate for design and fabrication uncertainties, including the refractive index, film thickness, etching depth, and waveguide width, to achieve phase matching for SH generation in the TFLN resonator. Different waveguide widths and sample temperatures can help us achieve wavelength tuning in the process of SH generation.

3. Second Harmonic Generation

3.1. Experimental Results

The setup shown in Figure 2 was utilized to characterize the SH signal in the RP double-layer TFLN microcavity. Tunable lasers in the 1550 and 780 nm bands were combined by a wavelength division multiplexer and coupled to the chip through a single-mode lensed fiber in the 1550 nm band. The power of the 1550 nm pump was controlled by a variable optical attenuation and monitored in real-time by a coupler and an optical power meter. The polarization controller is used to excite the intracavity $\text{TE}_{00, \text{FH}}$ mode. The pump and the generated SH signal are coupled out of the chip through another identical lensed fiber, separated by a wavelength division multiplexer, detected by their respective photodetector, and then fed into the oscilloscope. The 780 nm laser is mainly used to estimate the coupling loss of the signal and, on this basis, to calibrate the signal power monitored by the photodetector and oscilloscope. Assuming the two end-coupling efficiencies are the same, the fiber-to-chip losses of the pump and SH estimated based on transmittance are 8.4 and 8.0 dB/facet, respectively. The chip is placed on a stage with a thermoelectric

cooler and is temperature-adjusted to achieve double resonances of FH and SH to achieve maximum conversion efficiency. The microresonator in Figure 2 is a photograph that lights up due to the scattering of SH light.

The most efficient SH conversion was observed in a TFLN resonator with a 1.27 μm top width. Experimentally, pump transmission and SH signals were collected simultaneously while scanning the pump laser. Typical results are shown in Figure 3a, where the arrows in the transmission spectrum point to the $\text{TE}_{00, \text{FH}}$ mode, with a free spectral range (FSR) of 0.647 nm. The pump and signal modes that meet phase matching and energy conservation are located at 1518.4 and 759.2 nm, respectively. Optimizing polarization of pump light, the transmission spectrum of the pump and SH near the mode with the most efficient SH conversion was collected and is shown in Figure 3b. From Figure 3b, it is seen that the pump light is close to the critical coupling state, and the loaded and intrinsic quality factors of the TFLN resonator were measured to be 4.7×10^5 and 9.4×10^5 , respectively. When the on-chip power of the pump is 6.3 μW , the generated SH power was measured to be 154 nW, indicating a normalized (absolute) conversion efficiency of $3.9 \times 10^5 \text{ W}^{-1}$ (2.4%). Here, the normalized (absolute) SH conversion efficiency is defined as $\eta_{\text{norm}} = P_{\text{SH}}/P_{\text{FH}}^2 \cdot 100\%$ ($\eta = P_{\text{SH}}/P_{\text{FH}} \cdot 100\%$), with P_{FH} and P_{SH} indicating the pump and signal power on the chip, respectively.

We then changed the pump power and measured the SH conversion efficiency. The corresponding results are shown in Figure 3c, where the error mainly comes from the coupling between the waveguide and the lensed fiber. By fitting the experimental data according to the coupled mode theory equations,^[23] we obtained a normalized SH conversion efficiency of $4.4 \times 10^5 \text{ W}^{-1}$ under very weak pumping conditions. The inset of Figure 3c shows a linear fitting of the data collected with $P_{\text{FH}} < 14 \mu\text{W}$, which is in line with the theoretical conclusion, at low pump power, the output power of the SH is quadratic with the pump and indicated $\eta_{\text{norm}} = 3.5 \times 10^5 \text{ W}^{-1}$. The difference between $4.4 \times 10^5 \text{ W}^{-1}$, $3.9 \times 10^5 \text{ W}^{-1}$, and $3.5 \times 10^5 \text{ W}^{-1}$ is mainly due to more severe pump attenuation induced by the higher absolute SH conversion efficiency with increased pump power. We can see that when the on-chip pump power reaches

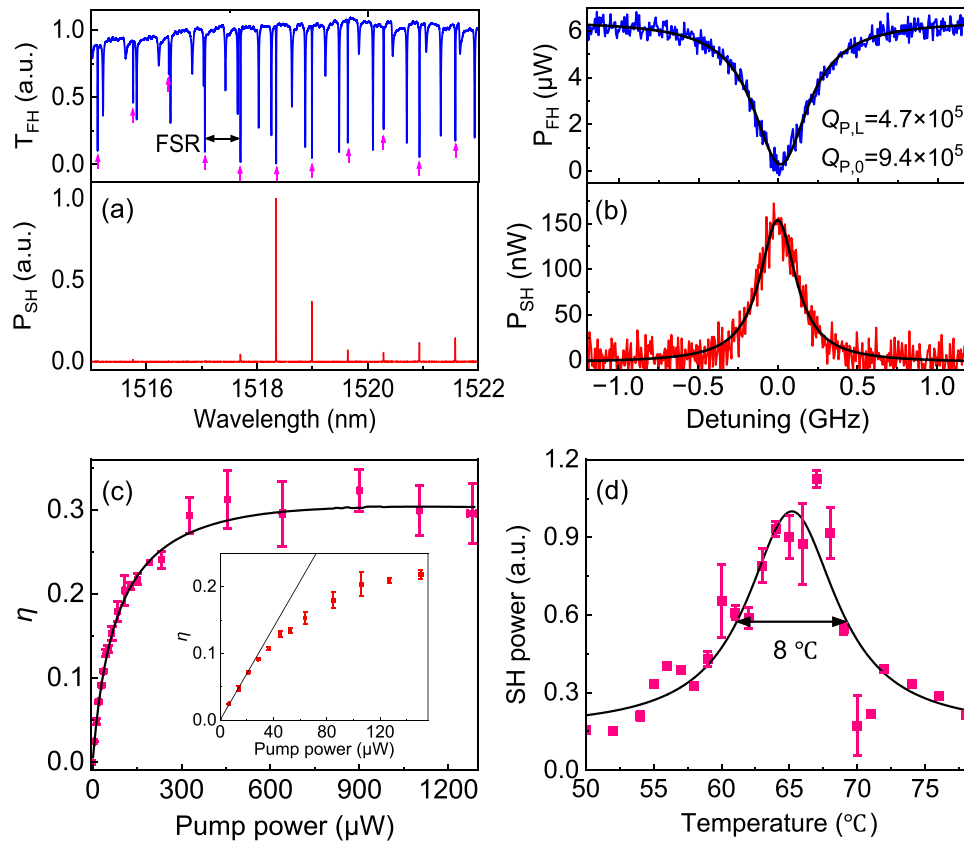


Figure 3. Second Harmonic Generation. a) The transmission spectrum of FH transmission and SH signal obtained by scanning the wavelength of the pump laser. b) The scanning signal of the piezo-controlled pump laser, the pump is basically in a critical coupling state, and the intrinsic Q value is above 9.4×10^5 . The conversion efficiency was calculated as $3.9 \times 10^5\% \text{ W}^{-1}$ under a 6.3 μ W pump. c) The SH conversion efficiency varies with the pump power. The inset shows the conversion efficiency with the low pump power. d) Temperature-tuned SH showing a bandwidth of 8 $^{\circ}$ C.

Table 1. Comparison of SH conversion efficiencies in microresonators.

Type	$\eta_{\text{norm}} [\% \text{ W}^{-1}]$	η_{max}
GaAs ^[22]	100	-
AlN ^[24]	1.7×10^4	11% @ 35 mW
LN ^[25]	1.5×10^3	-
LN ^[21]	9.9×10^3	-
PPLN ^[26]	2.3×10^5	6% @ 35.3 μ W
PPLN ^[27]	1.4×10^5	58% @ 3.4 mW
PPLN ^[19]	5.0×10^6	31% @ 14 μ W ^{a)}
LN [This work]	4.4×10^5	30% @ 600 μ W

^{a)} Estimated from Figure 3(f) in Reference.^[19]

600 μ W, the SH efficiency is close to saturation by 30%. We also measured the SH signal at different temperatures. The corresponding results are shown in Figure 3d, where the peak corresponds to the case that the FH and SH are dual-resonant. The bandwidth of the curve for the SH generation is 8 $^{\circ}$ C. The large temperature bandwidth makes the thermal-sensitive lithium niobate microcavity work stably.

In order to verify the SH mode in the microcavity, we use a 60 $\times$ microscope lens (NA = 0.85) and a visible CCD to monitor the mode profile at the ends of the single-pass waveguide with

a similar cross-section. We can clearly distinguish that the SH mode is $\text{TE}_{01, \text{SH}}$. At the pulley waveguide output facet, we can see that the waveguide mode related to the SH signal is $\text{TE}_{00, \text{SH}}$, which indicates the generated $\text{TE}_{01, \text{SH}}$ mode in the microring is coupled to the $\text{TE}_{00, \text{SH}}$ mode of the pulley waveguide.

3.2. Analysis and Discussion

According to the coupled mode equation, the maximum SH conversion efficiency depends on the coupling state of the pump and the signal,^[14]

$$\eta_{\text{max}} = \frac{Q_{\text{FH}, \text{L}}}{Q_{\text{FH}, \text{c}}} \cdot \frac{Q_{\text{SH}, \text{L}}}{Q_{\text{SH}, \text{c}}} \quad (5)$$

where $Q_{\text{FH}, \text{L}}$ ($Q_{\text{SH}, \text{L}}$) is the loaded quality factor of FH (SH), $Q_{\text{FH}, \text{c}}$ ($Q_{\text{SH}, \text{c}}$) is the quality factor of FH (SH) only considering the coupling loss. Because the pump is near critical coupling, we can infer that $Q_{\text{SH}, \text{L}}/Q_{\text{SH}, \text{c}} = 0.6$, and $\text{TE}_{01, \text{SH}}$ is also near the critical-coupled state. In the case of resonance and neglecting the pump attenuation, the normalized SH conversion efficiency^[14]

$$\eta_{\text{norm}} = \frac{64g^2}{\hbar\omega_{\text{FH}}^4} \frac{Q_{\text{SH}, \text{L}}^2}{Q_{\text{SH}, \text{c}}} \left(\frac{Q_{\text{FH}, \text{L}}^2}{Q_{\text{FH}, \text{c}}} \right)^2 \quad (6)$$

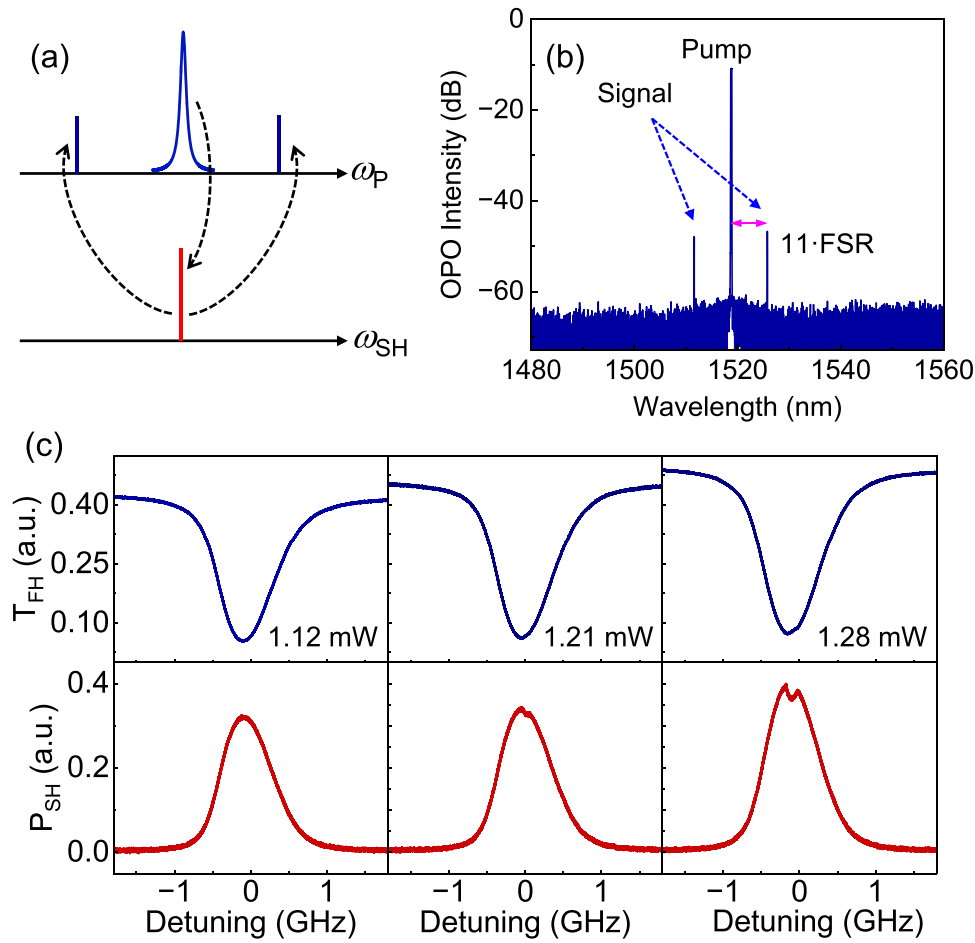


Figure 4. Cascaded optical parametric oscillation generation. a) Schematic diagram for cascaded OPO. b) The cascaded OPO signals at a pump of 1.28 mW. c) The transmission spectra of FH and SH at different pump powers.

where ω_{FH} is the angular frequency for the FH and $g = 6.35$ MHz is the nonlinear coupling strength when $(d_{eff})_{max} = 14.0$ pm V⁻¹ ($m = 1$) [see the [Supporting Information](#)], and we can infer that the $Q_{SH,L} = 1.3 \times 10^4$, $Q_{SH,c} = 2.2 \times 10^4$, and $Q_{SH,0} = 3.2 \times 10^4$. If $d_{eff} = 1.1$ pm V⁻¹ ($m = 3$), then calculate that $g = 0.5$ MHz, $Q_{SH,L} = 2.1 \times 10^6$. The $Q_{SH,L}$ of $TE_{01,SH}$ is greater than that of the fundamental mode, which is unreasonable due to the large mode loss of $TE_{01,SH}$. Therefore, we believe that we have achieved a quasi-phase matching (QPM) with $m = 1$.

The loaded quality factor of fundamental mode $TE_{00,SH}$ was measured to be 6×10^5 at 770 nm in the microring with $w_{ring} = 1.27$ μ m. Since the SH wavelength is 759.2 nm and outside the wavelength range (765–781 nm) of our tunable laser, we cannot directly excite the $TE_{01,SH}$ mode in the microring. Alternatively, we measured the quality factors of the microring with $w_{ring} = 1.21$ μ m on the same chip. The loaded Q of the $TE_{01,SH}$ mode was measured to be 1.4×10^4 , which is consistent with the SH generation experiments. The more significant loss of the $TE_{01,SH}$ mode is mainly due to the coupling of $TE_{01,SH}$ into other modes in the bending parts of the racetrack resonator (see the [Supporting Information](#)), which could be reduced by carefully designing deformed microring cavities.

Compared with the PPLN microcavities, the nonlinear coupling strength g of the double-layer microcavity is in the same order, but the intrinsic $Q_{01,SH}$ of the $TE_{01,SH}$ mode is lower, making the normalized conversion efficiency one order lower than that of the periodic poled microcavity reported.^[19] **Table 1** compares the most efficient SH in nanophotonic resonators. Our results are the best ones in resonators without PPLN domain structures.

4. Cascaded Optical Parametric Oscillation

In a higher pump power, we even observed cascaded optical parametric oscillation (OPO) using an optical spectrum analyzer. The principle of the cascaded OPO is shown in **Figure 4a**, where the pump (ω_p) drives the SH process and produces light with an angular frequency of ω_{SH} . A sufficiently high-power SH can drive the OPO process generating signals near the FH. The signal of the cascaded OPO obtained with a pump power of 1.28 mW is shown in **Figure 4b**, where the frequency of the OPO signal and the pump differ by 11 FSR.

The transmission spectrum of the FH and SH at different pump powers are shown in **Figure 4c**. There is a noticeable drop in the output power of the SH as a result of the SH light be converted back to the FH band,^[28] where we can estimate that the

threshold of the cascaded OPO is ≈ 1.21 mW. We calculate the theoretical threshold of cascaded OPO of 1.04 mW with experimental parameters, which is far less than the threshold of 350 mW for four-wave mixing with Kerr effects (see the [Supporting Information](#)). In addition, we noted that the pump loss caused by SH generation increased the pump transmittance, changed the pump from a critical coupling state to an undercoupling state, and reduces the pump Q . Therefore, the overcoupling state at low power will improve the maximum absolute conversion efficiency of SH generation. The low-threshold OPO has the potential to generate Pockels optical frequency combs for soliton generation and on-chip implementation of 1f–2f comb self-reference.^[1,28,29]

5. Conclusion

In summary, we have generated efficient SH conversion from $TE_{00, FH}$ to $TE_{01, SH}$ in an RP dual-layer TFLN microcavity. At an optimized temperature, the normalized SH conversion efficiency reaches $4.4 \times 10^5\% W^{-1}$ with weak pump power, and the absolute conversion efficiency is up to 30% when the pump power is 600 μW . At 1.21 mW pump power, cascaded optical parametric processes were also observed due to efficient SH generation. The simple preparation process and ultra-high conversion efficiency of the RP dual-layer TFLN microresonators will facilitate various applications, such as the frequency-up conversion of weak infrared signal and the generation of squeezed light and entangled photon pairs.

Note: In the final stages of revising the manuscript we became aware of a demonstration of high-efficiency second harmonic generation in a micro-resonator on dual-layered lithium niobate from F. Chen's group at Shandong University, China.^[30]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

microcavity, phase matching, second-harmonic generation, thin-film lithium niobate

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